

Labyrinthian user manual

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1. About the game

Labyrinthian is a 2D top-view pixel-art dungeon crawler style game. The goal of the game is to beat a set number of levels, each consisting of a randomly generated maze. The player must explore this maze to find the boss arena contained in the middle and defeat the boss of the maze to continue to the next level. Along his search the player may encounter enemies patrolling the labyrinth or clay pots, that can be broken to reveal various items which can aid the player.

2. Requirements and launch options

2.1 Requirements

To be able to play the game, the user must have the following installed on their computer:

- **Java runtime environment** version 21 or newer.
- **Maven** configured in the systems environment path.

2.2 Launch Options

The game can be launched from the terminal after navigating to the root of the project with the following command:

```
mvn javafx:run
```

This starts the game ready to play.

The project uses loggers which are turned off by default. Two levels of logging can be enabled.

1. INFO – informs about basic processes during game, enabled by starting the game with the following command:

```
mvn javafx:run -DlogLevel=INFO
```

2. DEBUG – shows detailed information about the states of entities. Enabled by launching the game with the following command:

```
mvn javafx:run -DlogLevel=DEBUG
```

3. Menu navigation

All buttons are controlled/pressed with a click of the left mouse button

3.1 Main Menu

Upon launching the game, the player is greeted with a main menu shown below:

Figure 1: Main Menu



- **Start New Game** – Starts a new playthrough from scratch. Does not rewrite saved game.
- **Load Game** – Loads and starts a saved game if save file is found.
- **Exit Game** – Closes the application.

3.2 Pause Menu

While the pause menu is shown, the game is frozen – entities don't move or attack. The in-game timer keeps running! If the player wishes to take a break from the game, it is recommended to save the game and exit to main menu.

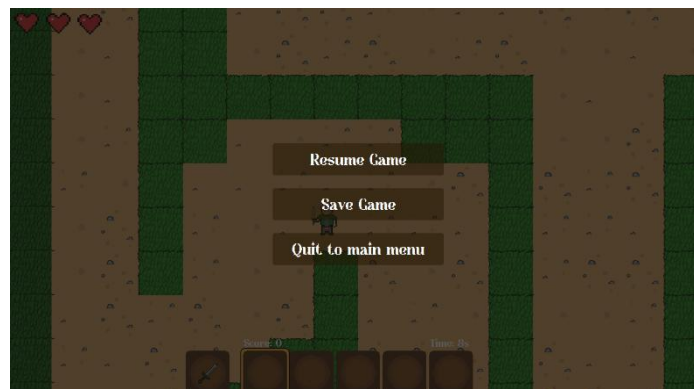


Figure 2: Pause Menu

- **Resume Game** – unfreezes the game and closes the pause menu.
- **Save Game** – saves the game, rewriting all previous saves.
- **Quit to main menu** – returns player to the Main Menu screen.

3.3 Dialogue menu / box

The dialogue box appears after the player interacts with an object or NPC. It either displays information – closable via the **Continue** button – or prompts the player to make a choice.



Figure 3: Closable dialogue

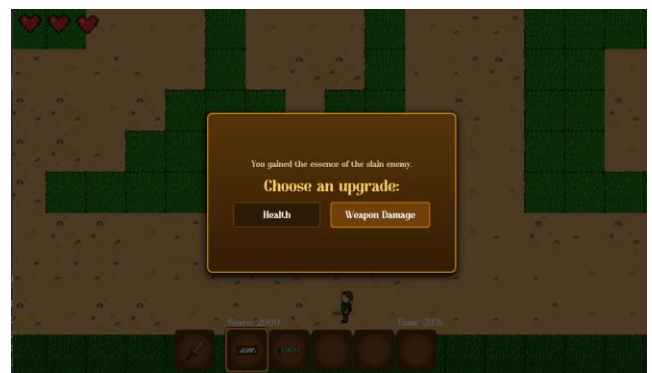


Figure 4: Multiple choice dialogue

3.4 End screens

3.4.1 Winning screen

Shown after the player beats 5 levels. Shows player stats

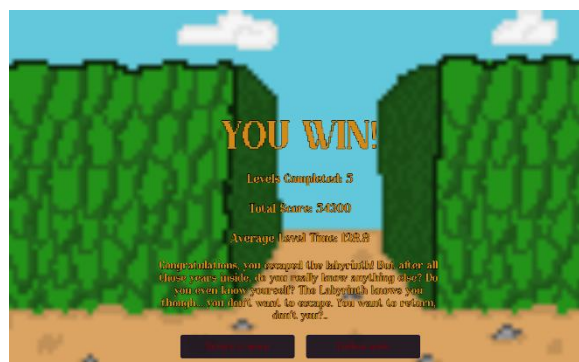


Figure 5: Winning screen

- **Return to menu** – returns player to the main menu screen.
- **Endless mode** – resumes the game in the “endless mode.”

3.4.2 Game Over Screen

Shown when player dies. Shows player stats.

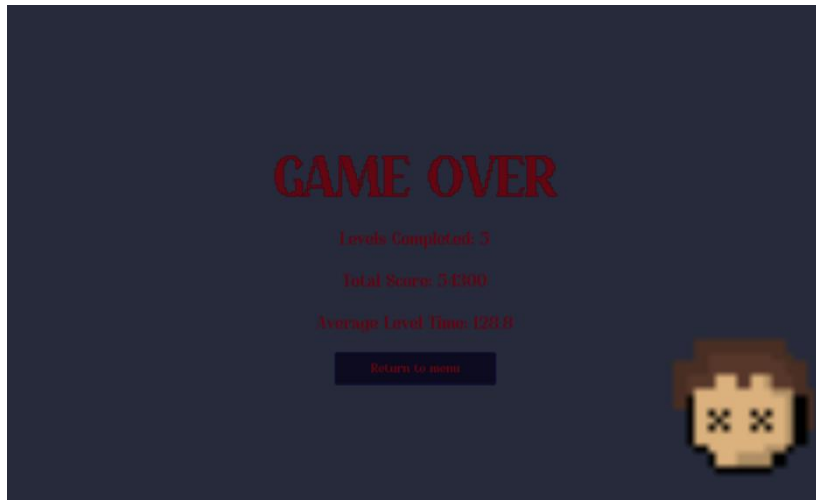


Figure 6: Game over Screen

- **Return to menu** – returns player to the main menu screen.

4. HUD information



Figure 7: HUD visible during gameplay

- **Health** – represented by hearts in the top left corner, each heart is 2 health points, half a heart is 1 health point.
- **Inventory** – represented by 6 squares in the center-bottom of the screen. Consists of 5 slots for items and a separate weapon slot
 - **Weapon slot** – Shows current weapon. Weapon is always active and cannot be deleted – player can attack no matter what other item they are currently holding
 - **Item slots** – Shows items picked up by player. The item is marked with a thick yellow border around the slot frame.
- **Score** – shows total concurrent score.
- **Time** – shows time elapsed since the start of the level.

5. Controls

The game uses keyboard inputs to control the game and mouse clicks to interact with menu buttons.

Movement:

- **W** – go up
- **S** – go down
- **A** – go left

- **D** – go right

Items and interactions:

- **1 – 5 (+ – ř0)** – sets the corresponding slot as active
- **E** – interaction (pick up nearby item on the ground, talk to NPC, transform BOSS)
- **Q** – Use currently active item
- **F** – Delete item in the current active slot
- **ESC** – Open the pause menu

Combat:

- **SPACEBAR** – attack in the direction the player model is facing. Must be pressed for each attack, cannot be held to keep attacking automatically.

6. Items

There are multiple clay pots around the map that the player can break with their attack. When destroyed the clay pots drop a random item that the player can pick up.

Figure 8: Claypot



6.1 Consumables

Items that the player can pick up into their item slots of the inventory and use.

Blinding stew



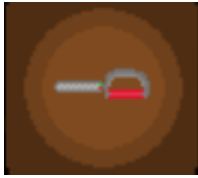
Using this item selects a random positive effect and applies it to the player also restricts their vision. All effects disappear after finishing the level.

- Add 3 hearts
- Double damage
- Double movement speed
- Activate life steal – i.e. player heals for their weapons damage after every hit



Figure 9: Blindness effect

CO2 Laser



If used when player is facing a hedge wall, converts that hedge wall into path. Cannot be used to convert walls of the boss arena. Has 3 uses.

CO2 Sharpened Shears



Same effect as laser but only 1 use.

Correct use of both items:



Figure 10a: facing the wall to be destroyed



Figure 10b: wall converted to path

Rusty POGO stick



Can be used to jump over a single hedge wall if the tile behind it is path. Cannot be used to jump into the boss arena. Infinite uses but deals 1 heart of damage and stuns the player on use.



Figure 11a: facing the wall to jump



Figure 11b: jumped over, stunned and took damage

SN-1C.K.E.R.S experimental bar



If player's health isn't full, use heals player for 2 hearts. If player is close enough to the boss, interaction attempts to transform the boss into an NPC. Consumes item regardless of success.

Yarn ball



When first used, starts drawing a red line behind where the player walks, indicating player's path. Second use deactivates the item. Subsequent reactivations start drawing new line and delete previous one.

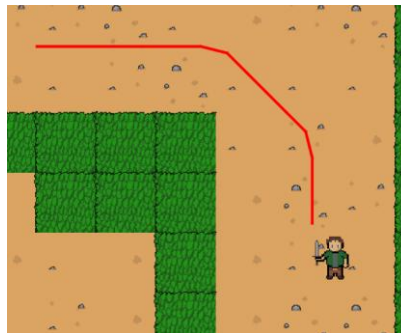


Figure 12: Line drawn behind player

6.2 Weapons

After pickup, weapons get equipped straight into the active weapon slot, replacing the previous weapon

Sword



The base weapon that the player starts with. At first has low damage but can be upgraded by killing enemies. Once replaced, cannot be found again.

Ultimate Obliterator



Rare weapon that can drop from a clay pot. After interaction prompts player to confirm that they want to equip the weapon. Weapon deals infinite damage, killing every enemy in one hit, but also permanently sets player health to 1 hp (half a heart), meaning player can always be killed with a single hit. Once the weapon is picked up, it cannot be replaced until the end of the run.

7. Gameplay mechanics and tips

In every level, the player spawns in the top-left corner of the maze. The goal is to locate the boss arena, which is always in the center with an entrance on the left side. The levels are randomly generated, meaning that finding the entrance can be relatively quick and easy or might take a while depending on the layout. Along the way, the player can encounter common enemies, clay pots, and ultimately the Boss. Defeating the boss spawns an exit portal that allows the player to proceed to the next level.

After beating 5 levels, the player is rewarded with a victory screen. In each subsequent level, enemies have more health and deal more damage, making the game progressively harder. The goal is to survive as long as possible while racking up as much score as possible.

7.1 Enemies

Each level contains 5 enemies that are spawned randomly in the maze, but always a set distance from player's starting point and never inside the boss arena.

Each enemy has a health bar above them indicating amount of health left.

The enemies wait in their spawn locations until a player gets too close. If player wanders within their detection range, enemies start chasing the player. If enemy gets close enough to player, it starts attacking them (indicated by "mad" sprite) dealing damage every couple seconds. If player gets far enough, it switches back to chasing mode. Once player escapes the enemy's vision for

long enough, or if the enemy gets too far from their spawn, it starts returning to its origin point, ignoring the player completely.

Once enemy is killed, the player is presented with the option to either add 1 heart to their health permanently or to upgrade their base weapon.

(Enemy remembers where it saw player last so hiding behind a corner might not be enough!)



Figure 13a: calm enemy



Figure 13b: attacking enemy

7.2 Clay pots

In each level there are 5 clay pots that randomly spawn in “dead ends” of the maze.

As already explained in the Items section, clay pots drop a random item when broken, that the player can pick up.

7.3 Boss

At the center of the arena stands the Boss, waiting for the player to find the path and the courage to face it. The boss has more health and deals more damage than the rest of the enemies and uses a different set of attacks.

7.3.1 Attacks

Once the player enters the arena, the boss starts chasing the player like a normal enemy would. When the player is yet too far, the boss uses its ranged attack: shooting a cone of 5 projectiles that spread out. Each projectile that hits the player deals damage, meaning if the player is too close when the Boss shoots, they can be hit by all projectiles at once and receive massive damage.

Once the boss gets close enough to the player, it switches to an AOE attack – the boss stops moving and a grey circle starts spreading from its center. Once it reaches the set range, the boss “jumps” and the circle fills with red color. If the player is standing within the circle when this happens, they take damage.



Figure 14: ranged attack

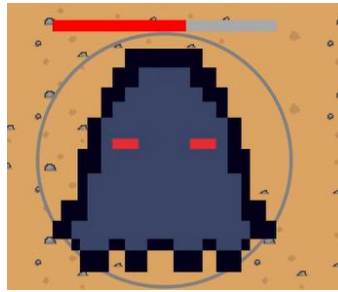


Figure 15a: AOE attack charging

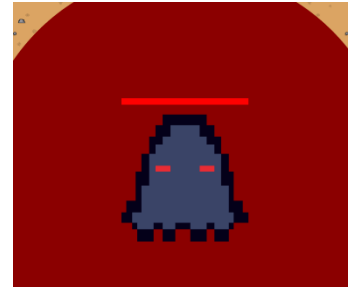


Figure 15b: AOE exploding

7.3.2 Defeat

Once the Boss is killed, it spawns an exit portal where the boss died. Interacting with the portal brings the player into the next level. A new map is generated, all enemies are respawned with more health and damage, and the player is teleported back to the starting position.



Figure 16: exit portal

7.3.3 Transformation

If the player is holding the **SN-1C.K.E.R.S bar** item in their active slot and interacts with the boss, there is a 10% chance to transform the boss into a friendly NPC. When the player interacts with the NPC, a thank you message is displayed and the exit portal spawns, allowing the player to safely continue to the next level.



Figure 17: NPC next to the portal

7.4 Scoring and strategy

The player can earn score in various ways, a possible goal for the player can be to amass as much as possible. Or the player can choose to try to reach as high of a level in the endless mode as they can. These two things don't necessarily correlate due to how the game is designed.

Events increasing player's score (n is the current level number):

- **Killing standard enemy** – grants $500 * n$ points
- **Killing the Boss** – grants $5\,000 * n$ points
- **Transforming the Boss** - grants $10\,000 * n$ points
- **Finishing the level under 150 s** – grants 50 points for every second saved.

Due to how the upgrades and scoring works and the randomness of the level, the player must carefully balance choosing the right upgrades at the right time as well as considering if sacrificing potential upgrades and items is worth the time saved during a level.